Capturing Hidden Semantics in Similarity Measures

- The above similarity measures cannot capture hidden semantics
  - Which pairs are more similar: Geometry, algebra, music, politics?
- The same bags of words may express rather different meanings
  - "The cat bites a mouse" vs. "The mouse bites a cat"
  - This is beyond what a vector space model can handle
- Moreover, objects can be composed of rather complex structures and connections (e.g., graphs and networks)
- New similarity measures needed to handle complex semantics
  - Ex. Distributive representation and representation learning